

Personalized Netflix Recommendation System

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This project presents a personalized recommendation system for Netflix-style content discovery. The system learns user preferences from historical movie ratings, predicts unseen ratings, compares multiple recommendation approaches, and generates Top-K movie recommendations using ranking-aware evaluation.

Project Package:

Complete source code, reproducible pipeline, dashboard, report, and presentation are included in the project package.

Introduction

Recommendation systems influence what users watch, buy, listen to, and discover. In this project, historical user-movie interactions are converted into personalized movie recommendations. The core challenge is to balance rating-prediction accuracy with the practical usefulness of the ranked recommendation list.

The final system is designed as a reproducible machine-learning pipeline: load data, analyze behavior, split data temporally, train multiple models, evaluate using both accuracy and ranking metrics, and generate Top-K recommendations.

Objectives

- Analyze user activity, movie popularity, rating behavior, temporal trends, and sparsity.
- Build models that predict unseen ratings and generate personalized Top-K recommendations.
- Compare baseline, item-based collaborative filtering, and matrix factorization methods.
- Evaluate the system using RMSE, MAE, MAP@10, Precision@10, Recall@10, and Hit Rate@10.
- Provide transparent recommendation examples and explanation logic.

Methodology

- **Temporal split:** older ratings are used for training and newer ratings are used for testing to reduce time leakage.
- **Bias baseline:** global mean plus user and movie bias terms.
- **Item-based collaborative filtering:** recommends movies similar to those the user has already rated highly.
- **Matrix factorization:** learns latent user and movie vectors with stochastic gradient descent.
- **Relevance definition:** for MAP@10, a movie is relevant when the actual rating is at least 3.5.

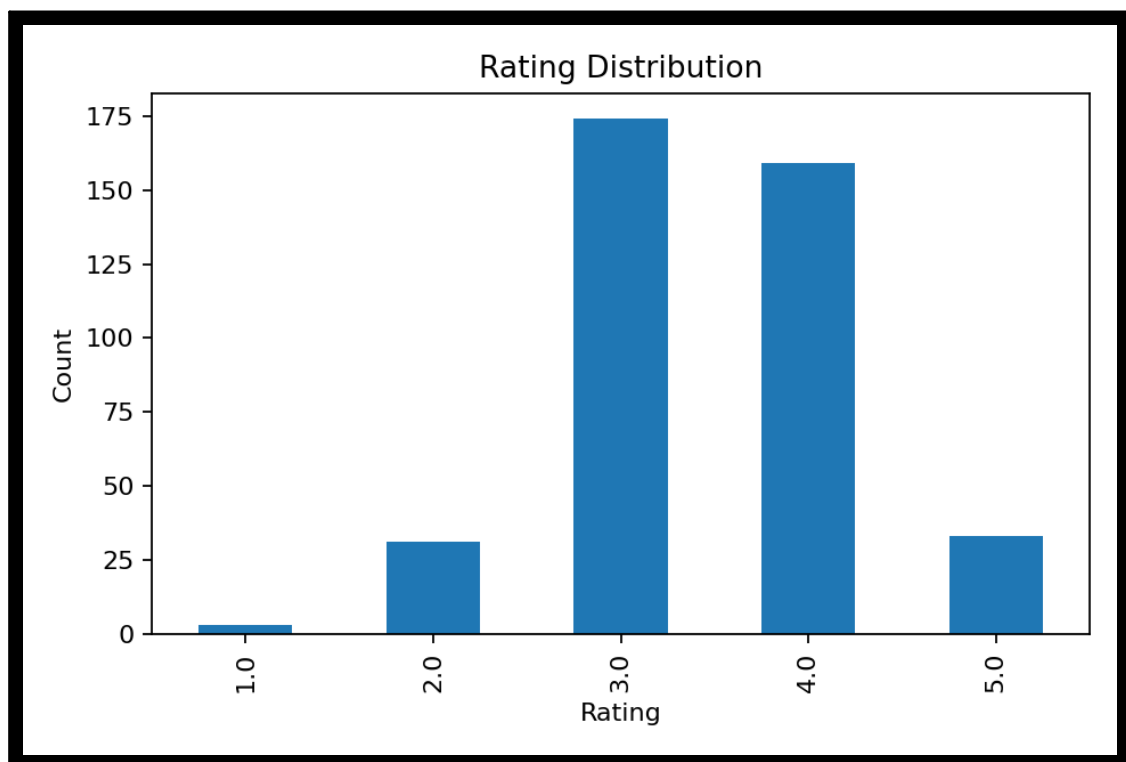
Dataset and Exploratory Data Analysis

The bundled demo uses a Netflix-style sample dataset so the project runs immediately on a normal laptop. The same pipeline supports the official Netflix Prize dataset after placing the Kaggle files in the raw data folder.

Metric	Value
Ratings	400
Users	40
Movies	18
Mean Rating	3.470
Sparsity	44.44%
Date Range	2003-01-21 to 2006-04-26

The dataset is sparse: users rate only a small part of the available movie catalog. This makes naive popularity recommendation insufficient and motivates collaborative filtering and latent-factor methods.

Rating Distribution

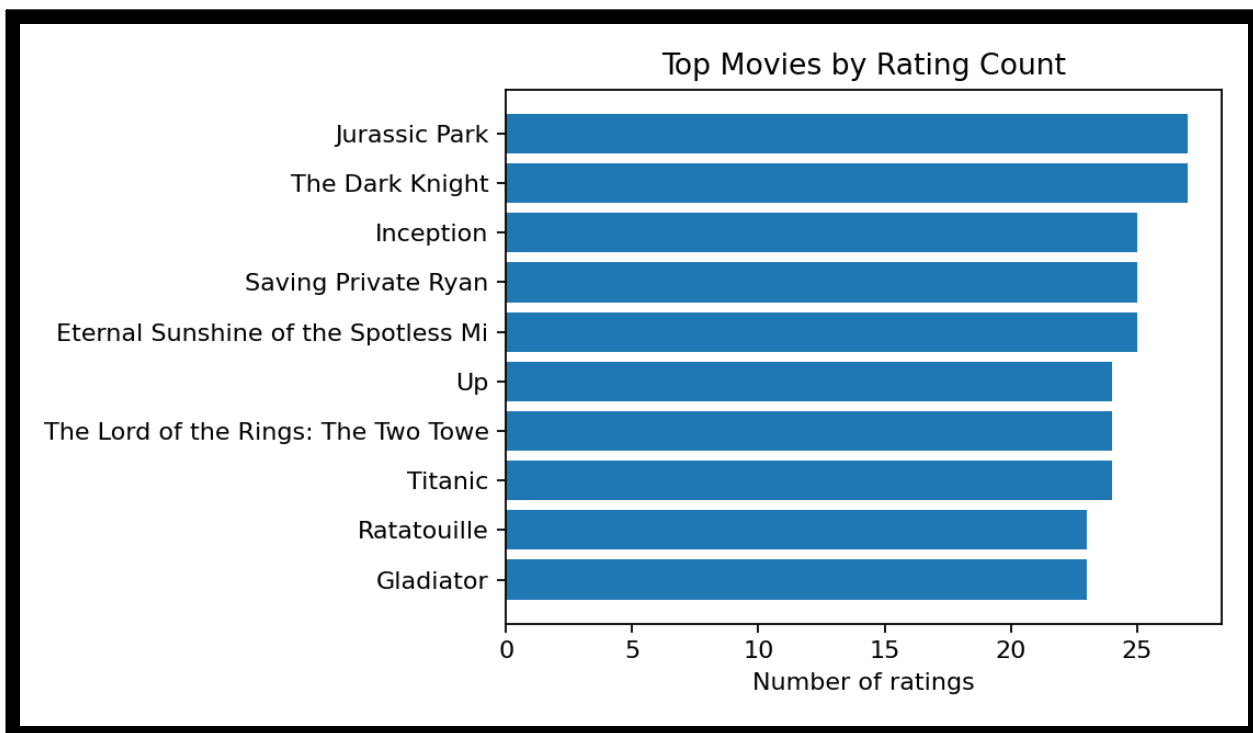


Most ratings are concentrated around 3 and 4 stars. This means a model can look good on RMSE by predicting average ratings, while still failing to rank the most useful movies at the top.

Data Visualization and Analysis

The visual analysis checks whether the system is dominated by active users, popular movies, or time-based rating bursts. These observations guide filtering, model choice, and evaluation design.

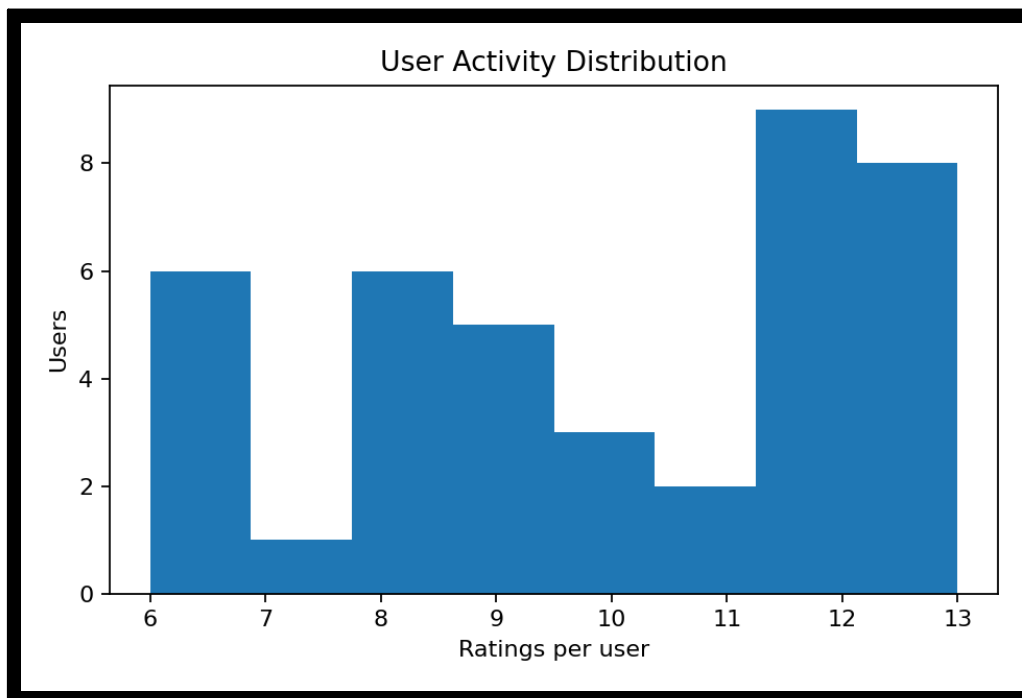
Top Movies by Rating Count



Popular movies such as Jurassic Park, The Dark Knight, and Inception receive high interaction volume. This creates a popularity-bias risk when the model has limited user history.

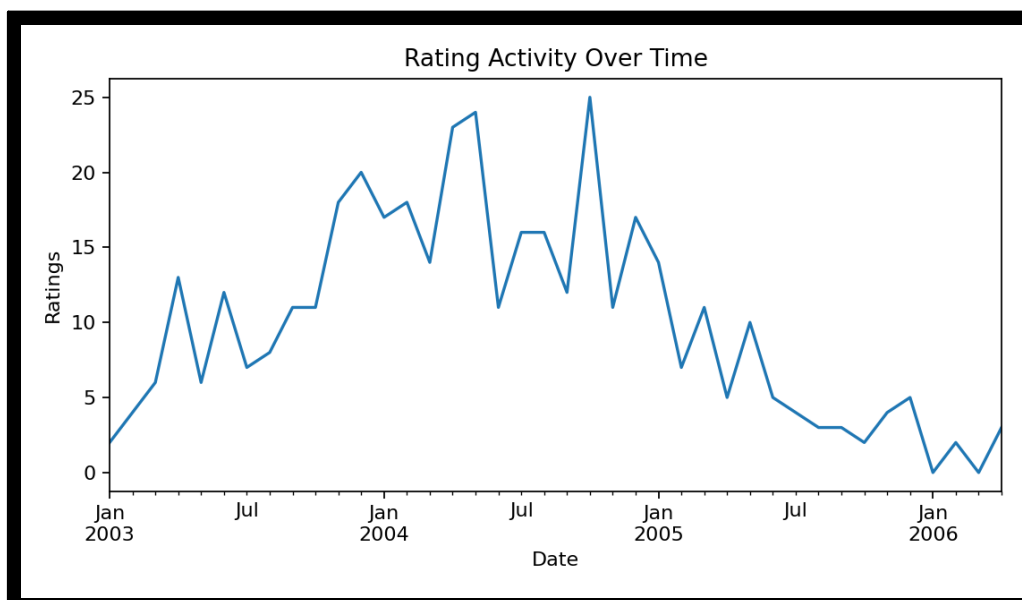
User and Temporal Activity

User Activity Distribution



Different users provide different amounts of rating history. Users with short histories are harder to personalize and are more vulnerable to generic recommendations.

Rating Activity Over Time



Rating activity changes across months, which supports the use of a time-based validation strategy instead of a purely random split.

Experimental Results

RMSE and MAE measure rating prediction accuracy. MAP@10, Precision@10, Recall@10, and Hit Rate@10 measure whether relevant held-out movies appear near the top of the recommendation list.

Model	RMSE	MAE	MAP@10	P@10	R@10	Hit@10	Users
item_item_cf	0.8136	0.6131	0.4188	0.1444	0.9630	1.0000	27
matrix_factorization_sgd	0.6442	0.5201	0.4162	0.1481	0.9815	1.0000	27
bias_baseline	0.6444	0.5534	0.3775	0.1481	0.9815	1.0000	27

On the bundled demo dataset, **item_item_cf** has the strongest ranking result with MAP@10 = 0.4188. The lowest RMSE is achieved by **matrix_factorization_sgd** with RMSE = 0.6442.

This result is important: the model with the best rating-prediction score is not always the model with the best ranking quality. For a real streaming platform, ranking quality is usually more valuable because the user mainly sees the top recommendations.

Performance Interpretation

- The bias baseline is a strong sanity check and should never be skipped.
- Item-based collaborative filtering gives the best MAP@10 and is easier to explain.
- Matrix factorization gives competitive RMSE but needs larger data and tuning to show its full advantage.
- The final competition report should regenerate metrics on the official Netflix Prize data or a clearly stated subset.

Recommendation Generation and Explainability

Top-K recommendations are generated by scoring movies the selected user has not already rated. The item-based model can also explain recommendations by showing which previously liked movies are most similar to the recommended title.

User	Rank	Movie	Title	Score
1	1	13	Eternal Sunshine of the Spotless Mind	5.000
1	2	2	The Shawshank Redemption	4.643
1	3	8	Gladiator	4.323
1	4	1	The Matrix	4.000
1	5	10	The Lord of the Rings: The Two Towers	4.000
1	6	9	Titanic	3.935
1	7	15	Ratatouille	3.914
1	8	18	Batman Begins	3.502

A user-facing explanation can be written as: "You may like this movie because it is similar to movies you rated highly." This makes the recommender more transparent than a black-box score alone.

Key Insights

- Sparse interactions make inactive-user and rare-item filtering important.
- Ranking metrics are essential because RMSE alone does not measure recommendation usefulness.
- Item-based collaborative filtering is practical for a dashboard prototype because it is simple, fast on small data, and explainable.
- Matrix factorization is promising for scale but requires more tuning and data volume.

Limitations and Future Improvements

Failure Cases and Limitations

- Cold-start users and new movies cannot be personalized without metadata or onboarding signals.
- Popularity bias can dominate when user history is short.
- The bundled demo metrics are not final competition metrics; official results must be regenerated on the Netflix Prize dataset or a declared subset.
- Large-scale training on the full Netflix data requires memory-aware batching, candidate sampling, and stronger compute.

Future Improvements

- Blend baseline, item-CF, and matrix-factorization scores in an ensemble.
- Add diversity re-ranking so the top recommendations are not too narrow.
- Use movie metadata for cold-start and explainable hybrid recommendations.
- Add batch scoring and sampled candidate evaluation for larger Netflix subsets.

Conclusion

The project delivers a complete recommendation-system pipeline with EDA, model comparison, RMSE, MAP@10, Top-K recommendations, explainability, and dashboard readiness. The design is simple enough to reproduce but strong enough to defend technically. The next serious step is scaling the same pipeline from the bundled demo dataset to the official Netflix Prize dataset or a clearly documented competition subset.